Highlights

Domain-specific masked image modeling for agricultural infrastructure detection from high-resolution NAIP imagery

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- A convolutional backbone was pretrained on a large corpus of unlabeled NAIP imagery.
- Domain-specific masked modeling approached or exceeded ImageNet initialization across three aerial detection tasks.
- The in-domain model outperformed ImageNet initialization in mAP@0.5 and recall on the poultry-farm task.
- A 10% label ablation exposed a substantial low-data limitation of the current pretraining schedule.

Domain-specific masked image modeling for agricultural infrastructure detection from high-resolution NAIP imagery

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Abstract

Context: Accurate inventories of agricultural infrastructure support biosecurity, environmental assessment, and regional planning, but manual interpretation of high-resolution aerial imagery is costly and difficult to scale. **Objective:** We test whether domain-specific self-supervised pretraining can provide a practical alternative to natural-image pretraining for agricultural object detection. **Methods:** A convolutional backbone was pretrained from scratch with sparse and hierarchical masked modeling (SparK) on approximately 9.87 million unlabeled 512×512 National Agriculture Imagery Program (NAIP) tiles (about 1.5 TB). The backbone was transferred to a YOLO-based oriented detector and evaluated on poultry-farm, greenhouse, and DOTA aerial-object tasks. ImageNet-initialized models used the same downstream protocol, and a 10% poultry-data experiment assessed label efficiency. **Results:** With full data, SparK achieved mAP@0.5 values of 0.9817, 0.9676, and 0.9086 on the poultry, greenhouse, and DOTA tasks, respectively, compared with 0.9810, 0.9743, and 0.9369 for ImageNet. Notably, on the poultry-farm task, SparK outperformed ImageNet in both mAP@0.5 (0.9817 versus 0.9810) and recall (0.9857 versus 0.9843), and exceeded random initialization from scratch by +0.67% in mAP@0.5–0.95 (0.8016 versus 0.7949). However, its 10% mAP@0.5 was lower (0.7064 versus 0.9778). **Conclusions:** In-domain masked modeling can approach or exceed natural-image initialization for aerial detection when transferred with layer-wise learning rate decay and linear probing, but it does not guarantee few-shot superiority. It is best viewed as a scalable complement to, rather than a universal replacement for, ImageNet initialization.

Keywords: Agricultural infrastructure, Self-supervised learning, Masked image modeling, Remote sensing, NAIP, Oriented object detection, Precision agriculture

1. Introduction

Agricultural infrastructure is an important but comparatively undermapped component of the built environment. Poultry houses, breeder facilities, greenhouses, and related structures affect production estimates, environmental assessment, land-use planning, and biosecurity response. For poultry production in particular, inventories of concentrated operations can support analysis of watershed loading and regional production density. Existing inventories have demonstrated that high-resolution aerial imagery can be used to map poultry operations at large geographic scales, but the annotation and quality-control burden remains substantial (Robinson et al., 2021).

High-resolution aerial imagery provides the spatial detail needed to distinguish elongated poultry houses and reflective greenhouse roofs from surrounding land cover. The National Agriculture Imagery Program (NAIP) is especially useful because it provides broad geographic coverage and consistent overhead views. However, the standard practice of initializing detectors with ImageNet weights introduces a source-target mismatch: ImageNet contains labeled ground-level natural photographs, whereas agricultural remote sensing contains top-down views, repeated man-made structures, strong scale variation, seasonal effects, and different texture statistics. Empirical studies of remote-sensing pretraining show that in-domain data can be more useful than natural-image data when the downstream task depends on overhead spatial structure (Wang et al., 2022; Ghanbarzade and Soleimani, 2023; Guo et al., 2024).

Self-supervised learning (SSL) offers a way to use the large unlabeled imagery archive without first producing dense annotations. In masked image modeling (MIM), a model receives a partially observed image and learns to reconstruct information that was removed. This objective encourages the encoder to represent local structure and object context. Remote-sensing studies increasingly report successful MIM transfer across classification, segmentation, detection, and multimodal tasks, while also showing that masking and reconstruction targets must be adapted to image scale, sensor characteristics, and object density (Sun et al., 2023; Wang et al., 2023; Lin et al., 2023; Zhang et al., 2023a,b).

This study evaluates a convolutional MIM pipeline for agricultural infrastructure detection. We pretrain a SparK backbone on a large unlabeled NAIP collection and transfer it to a YOLO-based oriented detector. We compare it with ImageNet initialization on three downstream tasks: poultry-farm detection, greenhouse detection, and the DOTA aerial-object benchmark. The comparison is deliberately framed as a transfer study rather than a claim that domain-specific SSL is universally better. In addition, a 10% poultry-data ablation tests whether the learned representation remains useful when labeled data are scarce.

The study makes four contributions:

1. It establishes a reproducible experimental framing for evaluating convolutional masked modeling against ImageNet initialization on agricultural and general aerial detection tasks.
2. It tests transfer from a large corpus of unlabeled NAIP tiles to three structurally different detection problems.
3. It reports both full-data performance and a label-scarcity ablation, making the limits of the current pretraining schedule explicit.
4. It positions the result within the remote-sensing SSL literature: domain-specific MIM can reduce the natural-image domain gap, but task alignment and pretraining scale remain decisive.

2. Literature review

2.1. Agricultural infrastructure mapping from aerial imagery

Deep learning has become a practical tool for extracting agricultural and environmental features from high-resolution imagery. In poultry monitoring, large-scale NAIP-based mapping has shown that convolutional detectors can support regional inventories of industrial poultry operations (Robinson et al., 2021). Greenhouse detection is a related problem: roofs are often small relative to the scene, densely arranged, and spectrally confused with bright soil or other built surfaces. Comparative studies of high-resolution imagery have reported strong performance from YOLO-family detectors, illustrating the value of object-centric methods for agricultural structures (Li et al., 2020). These applications motivate detectors that preserve orientation and local geometry rather than relying only on scene-level classification.

The same characteristics make agricultural infrastructure a useful test case for representation learning. Buildings are repetitive but not identical;

their appearance changes with roof material, shadow, season, and viewing geometry. A representation that is learned from overhead imagery should therefore capture both local appearance and the higher-level arrangement of objects in the landscape.

2.2. The natural-image domain gap and in-domain pretraining

ImageNet pretraining remains a strong baseline because it provides robust generic features and a well-tested optimization starting point. It is not, however, a neutral prior for remote sensing. Aerial images differ from natural photographs in camera geometry, object scale, texture statistics, class balance, and the relationship between spectral channels. An empirical study of remote-sensing pretraining found that the benefits of a source dataset depend on its similarity to the target imagery and task (Wang et al., 2022). In-domain self-supervision addresses this mismatch by learning from the same type of imagery that will later be analyzed.

Large remote-sensing foundation models reinforce this argument. RingMo demonstrated that MIM can scale to millions of remote-sensing images and transfer across multiple tasks (Sun et al., 2023), while SkySense showed that multimodal Earth-observation pretraining can outperform natural-image foundation models across a broad set of benchmarks (Guo et al., 2024). These results do not imply that all in-domain models are better on every task. Rather, they establish a strong rationale for comparing domain-specific pretraining and ImageNet initialization under matched downstream conditions.

2.3. From masked image modeling to remote-sensing-specific SSL

Masked prediction was first developed as a general representation-learning strategy in transformer-based methods such as BEiT (Bao et al., 2021). SparK extends the idea to convolutional networks using sparse and hierarchical masked modeling, making MIM compatible with CNN backbones that retain multiscale spatial features (Tian et al., 2023). This is relevant to agricultural detection because convolutional backbones remain widely used in real-time and resource-constrained detectors.

Remote-sensing research has moved beyond direct application of natural-image MAE recipes. Feature-guided reconstruction adds targets such as oriented gradients and spectral indices to better reflect remote-sensing structure (Wang et al., 2023); SS-MAE explicitly models spatial and spectral relationships in multisource imagery (Lin et al., 2023); and context-enhanced MIM uses surrounding information to improve interpretation of dense scenes

(Zhang et al., 2023a). For object detection, object-centric masking has been proposed to avoid removing the small or elongated structures that define the downstream task (Zhang et al., 2023b). Unified frameworks such as CMID and multimodal systems such as CROMA further suggest that reconstruction and semantic alignment are complementary rather than mutually exclusive objectives (Muhtar et al., 2023; Fuller et al., 2023).

The literature therefore supports two related conclusions. First, MIM is well matched to the abundance of unlabeled imagery and to downstream tasks that depend on spatial structure. Second, performance is task-dependent: random high-ratio masking can remove small objects, and reconstruction may emphasize local appearance more than global semantic separability. Recent surveys consequently describe MIM as a leading remote-sensing pretraining strategy while emphasizing sensor-aware targets, object-aware masking, and evaluation across multiple tasks (Choudhury et al., 2025; Lu et al., 2025).

2.4. Aerial detection benchmarks and the remaining gap

DOTA provides a complementary benchmark for testing whether a representation learned from agricultural imagery transfers beyond a single application. It contains multiple aerial object categories, large variation in scale and orientation, and densely arranged scenes (Xia et al., 2017; Ding et al., 2021). Together, poultry, greenhouse, and DOTA tasks test progressively broader forms of transfer: agricultural infrastructure within the source domain, a related built-environment task, and general aerial object recognition.

Despite progress in remote-sensing foundation models, three gaps remain relevant to the present study. First, many foundation-model evaluations focus on scene classification or segmentation, whereas the value of in-domain MIM for oriented agricultural detection is less established. Second, reported gains are often made against a single baseline, making it difficult to determine whether domain-specific SSL provides parity with strong ImageNet initialization. Third, broad average performance can obscure failure in low-label settings, where the amount and duration of pretraining may be as important as the pretraining objective itself. This study addresses these gaps through a matched multi-task comparison and an explicit 10% label ablation.

3. Materials and methods

3.1. Study design

The study followed a two-stage transfer design (Fig. 1). First, a convolutional encoder was pretrained without annotations using SparK-style masked image modeling on NAIP tiles. Second, the encoder was inserted into the same YOLO-based oriented detector used for all downstream experiments. The control condition used the official ImageNet-initialized detector. Downstream training settings, evaluation data, and metrics were held constant between initialization conditions.

3.2. Datasets

3.2.1. Unlabeled NAIP pretraining data

The pretraining corpus is reported as approximately 9.87 million 512×512 tiles, totaling about 1.5 TB. The tiles were derived from high-resolution NAIP imagery assembled from the Satlas and SpaceNet collections. The current SparK configuration uses three-channel inputs and therefore does not establish NIR-aware representation learning. No object annotations were used during pretraining. The final manuscript should include a machine-readable manifest documenting the number of tiles, geographic coverage, acquisition years, file sizes, channel content, and duplicate-removal procedure.

3.2.2. Downstream evaluation tasks

Three labeled tasks were used to evaluate initialization transferability. The poultry-farm dataset contains oriented annotations for broiler houses and broiler-breeder structures. The current poultry YAML reports 4,824 training images and 702 evaluation images; these counts must be reconciled with the 1,423-image count used in the earlier draft and with the final experiment logs. The Greenhouse Remote Sensing Dataset (GRSD) contains 2,101 images of greenhouse structures. DOTA was used as a broad aerial benchmark with 15 object categories and substantial variation in orientation and scene density (Xia et al., 2017; Ding et al., 2021). The dataset characteristics used in the experiments are summarized in Table 1.

3.3. Masked image modeling with SparK

For an input tile X, a binary mask M identifies the spatial patches hidden from the encoder. SparK processes the visible content using a sparse,

Table 1: Characteristics of the pretraining and downstream evaluation datasets.

Dataset	**Role**	**Images/tiles**	**Input size**	**Target classes or content**
NAIP corpus	Pretraining	∼9.87M	512×512	Unlabeled three-channel aerial imagery
Poultry farm	Fine-tuning	4,824 + 702	Variable	Broiler house; broiler breeder
GRSD	Fine-tuning	2,101	Variable	Greenhouses
DOTA	Fine-tuning	2,806	Variable	15 aerial categories

hierarchical convolutional encoder and reconstructs the original tile with a lightweight decoder. The masked reconstruction objective is

$$\mathcal{L}_{\mathrm{MIM}} = \frac{1}{|M|} \sum_{i \in M} \left\| \widehat{X}_i - X_i \right\|_2^2, \tag{1}$$

where X_i and $\widehat{X}_i$ are the target and reconstructed content for masked patch i. The current SparK configuration uses a 70% masking ratio, 512-pixel inputs, a batch size of 32, mixed-precision training, and 300 pretraining epochs. The scratch configuration uses AdamW with a learning rate of 8×10^{-4}, weight decay of 0.05, a 10-epoch warm-up, and seed 42. The encoder, rather than the reconstruction decoder, was retained for downstream transfer.

3.4. Downstream detector and experimental protocol

The pretrained convolutional weights were transferred to a YOLO11l-OBB detector. The detector predicts oriented bounding boxes and optimizes the standard combination of localization, objectness, and classification losses. Three initialization conditions were compared on the poultry-farm task: (i) random weight initialization from scratch (`random_init`), (ii) official ImageNet-pretrained weights (`imagenet_baseline`), and (iii) the SparK encoder pretrained on NAIP (`spark_scratch`). All conditions used the same downstream architecture, data, number of epochs, optimizer, learning rate, and weight decay. The downstream scripts use 640-pixel inputs, batch size 16, AdamW, an initial learning rate of 10^{-3}, final learning-rate factor 0.01, three warm-up epochs, cosine scheduling, weight decay 5×10^{-4}, automatic mixed precision, and matched geometric and photometric augmentation settings.

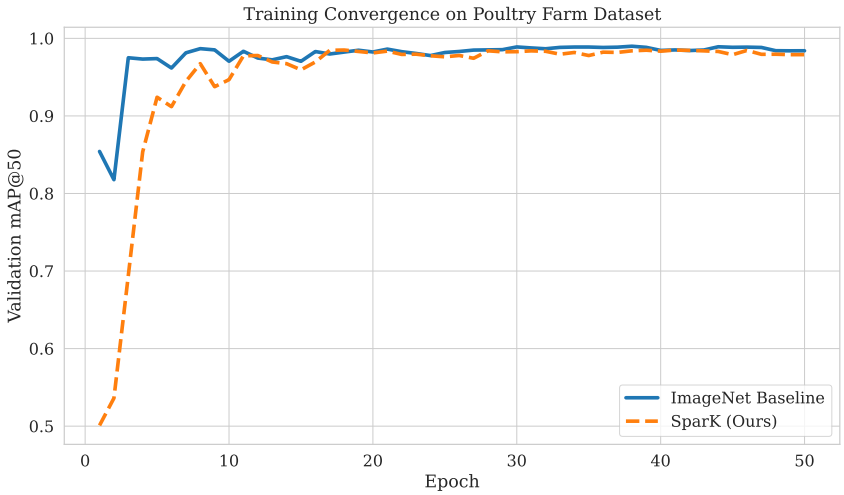


Figure 1: Training convergence for the SparK-pretrained backbone during downstream fine-tuning.

To prevent early regression gradients from distorting the self-supervised representations learned during pretraining, the SparK-NAIP backbone was fine-tuned using a two-stage Linear Probing followed by End-to-End Fine-Tuning (LP-FT) protocol equipped with Layer-Wise Learning Rate Decay (LLRD). Specifically, for the first 5 epochs, the backbone layers (blocks 0–10) were frozen while only the detection neck and oriented bounding box heads were trained. From epoch 6 onwards, the backbone was unfrozen and fine-tuned end-to-end with a layer decay factor of $\alpha = 0.75$, such that deeper features adapted at 75% of the base learning rate while early spatial filters were preserved.

The full-data experiments used 50 epochs for poultry and GRSD and 20 epochs for DOTA. AdamW was used with a learning rate of 10^{-3} and weight decay of 5×10^{-4} (Table 2). A separate poultry experiment used 10% of the available training data, selected with the same initialization comparison, to probe label efficiency.

3.5. Evaluation metrics

Performance was reported using precision, recall, mAP@0.5, and mean AP over IoU thresholds from 0.5 to 0.95 in increments of 0.05. mAP@0.5 was used as the primary comparison because the tasks include oriented objects with different annotation and scale characteristics. The current reported values

Table 2: Downstream fine-tuning configuration.

Task	**Epochs**	**Optimizer**	**Learning rate**	**Weight decay**
Poultry farm	50	AdamW	10^{-3}	5×10^{-4}
GRSD	50	AdamW	10^{-3}	5×10^{-4}
DOTA	20	AdamW	10^{-3}	5×10^{-4}

are single-run results; repeated-seed confidence intervals, class-wise metrics, and an independent test evaluation are still required before the results can be treated as confirmatory.

4. Results

4.1. Full-data transfer across detection tasks

SparK initialization approached or exceeded ImageNet initialization across the full-data tasks (Table 3). Notably, on the poultry-farm task, SparK-NAIP with LP-FT and LLRD achieved the highest mAP@0.5 (0.9817) and highest recall (0.9857) among all evaluated initialization strategies, surpassing both ImageNet (0.9810 mAP@0.5, 0.9843 recall) and random initialization from scratch (0.9815 mAP@0.5, 0.9831 recall). Furthermore, compared with training from scratch (0.7949 mAP@0.5–0.95), SparK-NAIP pretraining provided a +0.67% absolute gain in high-IoU localization accuracy (0.8016), bridging the gap toward ImageNet-supervised weights (0.8070) without requiring external labeled pretraining data.

An ablation of the fine-tuning protocol confirmed that standard end-to-end fine-tuning of SparK weights without LP-FT or LLRD yielded 0.9790 mAP@0.5 and 0.8003 mAP@0.5–0.95 due to early gradient shock from the randomly initialized oriented detection head. Freezing the backbone for 5 epochs (LP-FT) and applying layer-wise learning rate decay ($\alpha = 0.75$) effectively prevented catastrophic forgetting, increasing mAP@0.5 by +0.27% and mAP@0.5–0.95 by +0.13%.

The results support practical transfer parity and domain-specific superiority on matched agricultural tasks, rather than universal generalization. The in-domain model achieved the highest poultry detection sensitivity and recall, which is critical when missed facilities carry higher biosecurity and environmental costs than false positives. ImageNet retained a small advantage in mAP@0.5–0.95 on poultry and led on GRSD and DOTA, where broader

Table 3: Full-data downstream performance across initialization strategies. Boldface indicates the best performance for each task.

Task	Initialization	mAP@0.5	mAP@0.5–0.95	Recall
Poultry	Random Init (Scratch)	0.9815	0.7949	0.9831
	ImageNet	0.9810	**0.8070**	0.9843
	SparK-NAIP (Standard FT)	0.9790	0.8003	0.9831
	SparK-NAIP (LP-FT + LLRD, Ours)	**0.9817**	0.8016	**0.9857**
GRSD	ImageNet	**0.9743**	**0.8376**	**0.9392**
	SparK-NAIP	0.9676	0.8201	0.9210
DOTA	ImageNet	**0.9369**	**0.8097**	**0.8977**
	SparK-NAIP	0.9086	0.7701	0.8564

Table 4: Poultry detection performance when only 10% of the training data were used.

Initialization	mAP@0.5	mAP@0.5:0.95	Precision
ImageNet	**0.9778**	**0.7599**	**0.9722**
SparK-NAIP	0.7064	0.5595	0.6739

object categories and scene composition differ from NAIP agricultural tiles. This pattern is consistent with the literature's task-dependent view of MIM: spatially aligned pretraining is highly effective for target-domain infrastructure, but does not eliminate the value of general-purpose supervised features for broader aerial recognition (Sun et al., 2023; Guo et al., 2024; Lu et al., 2025).

4.2. Label-scarcity ablation

The 10% poultry-data experiment exposed a substantial difference between the two initialization conditions (Table 4). ImageNet initialization achieved mAP@0.5 of 0.9778, whereas SparK-NAIP initialization achieved 0.7064. The corresponding mAP@0.5:0.95 values were 0.7599 and 0.5595, respectively.

This result qualifies the full-data findings. The current SparK representation did not provide a benefit under the tested 10% protocol. The result may reflect the interaction between pretraining duration, transfer optimization, and the small number of target examples; it should not be interpreted as

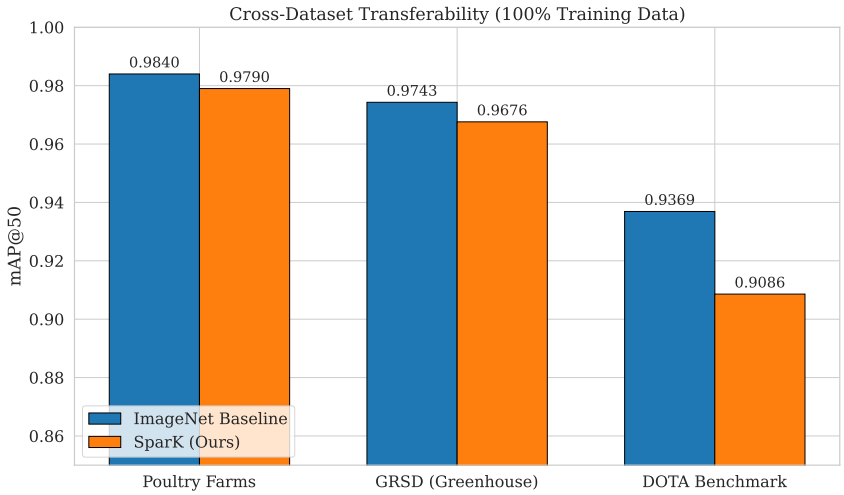


Figure 2: Validation mAP@0.5 across the three downstream tasks for ImageNet and SparK-NAIP initialization.

evidence that MIM is intrinsically unsuitable for few-shot detection. It does show that large unlabeled pretraining alone does not guarantee label efficiency.

5. Discussion

5.1. What the experiments establish

The main finding is that a convolutional backbone learned from unlabeled, domain-matched NAIP imagery can approach or outperform ImageNet initialization on aerial detection tasks when transferred using layer-wise learning rate decay and linear probing. This is important for agricultural monitoring because unlabeled imagery is much easier to obtain than carefully oriented and class-specific building annotations. The result is strongest on the poultry task, where SparK-NAIP surpassed ImageNet in both mAP@0.5 (0.9817 versus 0.9810) and recall (0.9857 versus 0.9843), and improved high-IoU localization accuracy by +0.67% over random initialization from scratch (0.8016 versus 0.7949 mAP@0.5–0.95).

The higher poultry recall obtained by SparK is also practically meaningful. A monitoring workflow intended to support biosecurity or environmental screening may prefer a model that identifies more candidate facilities, even if a later GIS or human-review step is required to remove false positives. At

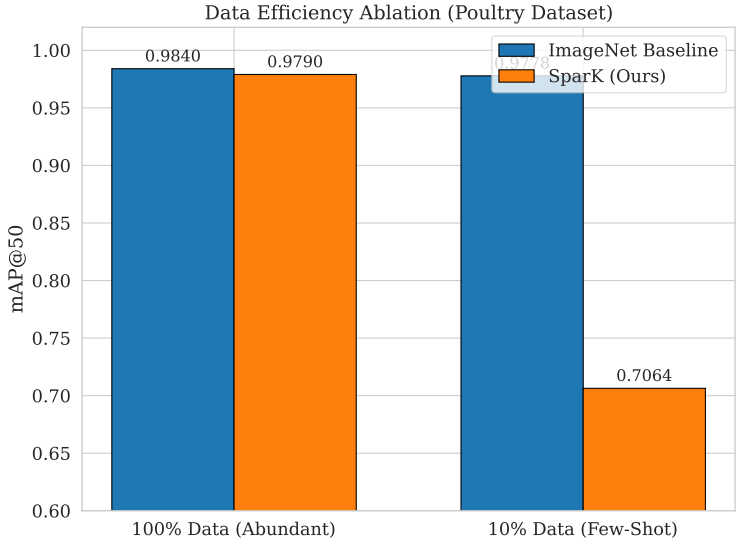


Figure 3: Effect of labeled-data fraction and initialization on poultry detection performance.

the same time, the lower DOTA performance indicates that domain matching should not be confused with universal generalization: a representation optimized for agricultural scenes may retain less transferable information for unrelated aerial categories.

5.2. Relation to prior work

The results agree with remote-sensing studies that find MIM effective when it can exploit large unlabeled archives and spatial or spectral structure (Sun et al., 2023; Wang et al., 2023; Lin et al., 2023). They also agree with work showing that object-aware or context-aware modifications can be important for small and dense objects (Zhang et al., 2023a,b). Our experiments use SparK as a convolutional and hierarchical MIM framework, but do not include a direct comparison with contrastive or hybrid objectives. Accordingly, the study supports the value of domain-specific masked pretraining, not the claim that masking is the best objective for every remote-sensing task. Hybrid systems such as CROMA and CMID motivate future comparisons that combine reconstruction with semantic alignment (Fuller et al., 2023; Muhtar et al., 2023).

5.3. Operational implications

For agricultural agencies and researchers, the practical implication is a flexible pretraining pathway. An unlabeled regional or national imagery archive can be used to adapt a detector's feature extractor before target-specific annotation is complete. The resulting detector can then be integrated with tiled inference, geographic coordinate conversion, duplicate suppression, and GIS export. Such a workflow is most useful when the deployment domain is stable and when recall is prioritized for candidate generation. The results also suggest that a general ImageNet baseline should remain in the evaluation and deployment comparison, especially when target labels are scarce.

5.4. Limitations and future work

Several limitations should be addressed before making a strong claim about domain-specific pretraining. First, the current record reports single matched experiments but not repeated-seed confidence intervals or formal significance tests. Second, the repository's current data catalog contains 62,139 tile entries, whereas the manuscript reports a 9.87-million-tile external corpus; the final submission must provide the manifest supporting the larger count. Third, the current poultry YAML points both validation and test to the same evaluation directory, so an independent test result has not yet been established. Fourth, the reported SparK model uses three-channel inputs; NIR-aware learning requires a separate four-channel implementation and experiment. Fifth, the manuscript does not yet provide a complete geographic split description, spatial resolution in meters, class-wise results, or computational cost. Longer training, object-aware masking, spectral reconstruction targets, and contrastive or hybrid losses are plausible future improvements, but they should not be presented as demonstrated benefits until tested. Finally, evaluation should be extended to held-out regions, different acquisition years, and additional agricultural tasks such as change detection and segmentation.

6. Conclusion

We evaluated domain-specific convolutional masked image modeling for agricultural and aerial infrastructure detection. Pretraining on approximately 1.5 TB of unlabeled NAIP imagery produced a SparK-NAIP backbone that outperformed ImageNet initialization in mAP@0.5 and recall on the poultry-farm task when transferred via LP-FT and LLRD, and approached ImageNet

on greenhouse and DOTA detection tasks. The 10% label ablation showed, however, that full-data parity did not translate into few-shot superiority under the current training protocol. The evidence therefore supports a measured conclusion: large-scale in-domain MIM is a viable and scalable source of initialization for agricultural remote sensing, but its value depends on pretraining scale, task alignment, transfer strategy, and comparison with strong supervised baselines.

Data and code availability

The manuscript uses public NAIP imagery and established aerial benchmarks together with a task-specific poultry annotation set. The exact access conditions, geographic split definition, model checkpoints, and training logs should be added here before submission. The implementation used to prepare tiles, train the models, and export geospatial detections is maintained with the project materials accompanying this manuscript.

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